

# Analytic Number Theory

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# Notation

This section summarizes the main symbols and notations used throughout the thesis.

| Symbol                              | Meaning                                       |
|-------------------------------------|-----------------------------------------------|
| $d n$                               | d is a divisor of n                           |
| $\lfloor \cdot \rfloor$             | floor function                                |
| $f = O(g)$                          | f is bounded by g, i.e. $ f(x)  \leq Cg(x)$   |
| $f \sim g$                          | $\frac{f(x)}{g(x)} \rightarrow 1$             |
| $n!$                                | $\prod_{1 \leq k \leq n} k$                   |
| $\binom{a}{b}$                      | binomial, i.e. $\frac{a!}{(a-b)!b!}$          |
| $(m, n)$                            | greatest common divisor (gcd) of $m, n$       |
| $[m, n]$                            | least common multiple (lcm) of $m, n$         |
| $a \equiv b \pmod{q}$               | a and b are congruent modulo q                |
| $f : x \mapsto f(x)$                | define function $f$ and variable $x$          |
| $\mathbb{Z}^+$ or $(\mathbb{Z}, +)$ | group of integer set under addition operation |

# Function

This section summarizes the main symbols and notations used throughout the thesis.

| Symbol       | Meaning                   |
|--------------|---------------------------|
| $\pi(x)$     | prime counting function   |
| $\zeta(s)$   | zeta function             |
| $\Lambda(n)$ | von Mangoldt function     |
| $M_f(x)$     | summation function of $f$ |
| $\mu$        | Möbius function           |
| $L(s, f)$    | Dirichlet Series          |
| $\varphi$    | Euler function            |

# Mathematics: How it Works

Mathematics is like building a castle from bricks. We usually start from *definition*, then prove some *proposition* and *lemma*, and finally prove the central results *theorem* and its consequence *corollary*.

| Type        | Purpose                                              | Typical Usage                                             |
|-------------|------------------------------------------------------|-----------------------------------------------------------|
| Definition  | Introduces a new concept, object, or notation.       | Used when explaining what something <i>is</i> .           |
| Proposition | A less central but still important result.           | Often a useful or supporting fact.                        |
| Lemma       | A smaller, helper result used to prove a theorem.    | Usually not important by itself, but essential in proofs. |
| Theorem     | A major mathematical statement that has been proved. | Central results of a section or paper.                    |
| Corollary   | A direct consequence of a theorem or proposition.    | Follows almost immediately from a previous result.        |

# 1 Counting Prime Numbers

## 1.1 Introduction

It has been known since the time of Euclid that there are infinitely many prime numbers. Arguing by contradiction, suppose that there were only finitely many primes  $p_1, \dots, p_n$ . Then the number  $p_1 \cdots p_n + 1$  must have a prime divisor not equal to any of  $p_1, \dots, p_n$ . In this course we will be interested in quantifying the infinitude of prime numbers. To do so, we define the prime counting function

$$\pi(x) = \#\{p \in \mathcal{P} : p \leq x\}.$$

Euclid's theorem therefore says that  $\pi(x) \rightarrow \infty$  as  $x \rightarrow \infty$ , but the question is

*at what rate?*

**Theorem 1.1** (Prime Number Theorem (PNT)). *As  $x \rightarrow \infty$  we have*

$$\pi(x) \sim \frac{x}{\log x}.$$

## 1.2 Euler's Method

**Zeta Function:** For  $s > 1$  one considers the convergent series

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}.$$

In terms of prime numbers:

$$\zeta_p(s) = 1 + \frac{1}{p^s} + \frac{1}{(p^2)^s} + \cdots + \frac{1}{(p^\alpha)^s} + \dots$$

As geometric series, we have

$$\zeta_p(s) = (1 - 1/p^s)^{-1}$$

**Key observation:**

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s} = \prod_p \zeta_p(s) = \prod_p (1 - 1/p^s)^{-1} \tag{1}$$

**Theorem 1.2** (Fundamental Theorem of Arithmetic). *Every integer  $n \geq 2$  can be expressed as a product of prime numbers. Moreover, this factorization is unique up to the order of the factors. That is, there exist distinct prime numbers  $p_1, p_2, \dots, p_k$  and positive integers  $\alpha_1, \alpha_2, \dots, \alpha_k$  such that*

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}.$$

Assume there are finite primes,  $\prod_p (1 - 1/p^s)^{-1}$  will be finite, but the  $\sum_{n \geq 1} \frac{1}{n^s}$  will be infinite as  $s \rightarrow 1$ , which contradicts our assumption, thus the number of primes is infinite.

If we take logarithm on each side, the equation can be written as,

$$\log \zeta(s) = \log \prod_p (1 - 1/p^s)^{-1} = - \sum_p \log(1 - 1/p^s) \approx \sum_p 1/p^s \quad (2)$$

The  $\zeta(s)$  is infinite as  $s \rightarrow 1$ , thus the series

$$\sum_p 1/p \quad (3)$$

is *divergent*.

### 1.3 Chebyshev's Method

**Theorem 1.3.** *There exist constants  $0 < c < C$  such that for  $x \geq 2$  one has*

$$c \frac{x}{\log x} \leq \pi(x) \leq C \frac{x}{\log x}.$$

**Definition 1.4** ( $p$ -adic valuation). *For  $n \in \mathbb{Z} \setminus \{0\}$  and  $p$  a prime number, the  $p$ -adic valuation of  $n$ , written  $v_p(n)$ , is the largest integer  $\alpha \geq 0$  such that  $p^\alpha$  divides  $n$ . That is to say, such that  $p^\alpha \mid n$  and  $p^{\alpha+1} \nmid n$ . In particular, one has*

$$n = \prod_{p \mid n} p^{v_p(n)} = \prod_{p \in \mathcal{P}} p^{v_p(n)}.$$

Define  $\theta(x)$ :

$$\theta(x) = \sum_{p \leq x} \log p$$

**Theorem 1.5** (Mertens). *We have*

$$\sum_{p \leq x} \frac{\log p}{p} = \log x + O(1).$$

*Proof.* **Key Observation:**

$$n! = \prod_{p \leq n} p^{v_p(n!)} \quad (4)$$

$$\log(n!) = \sum_{p \leq n} v_p(n!) \log p$$

The left hand side can be written as,

$$\log(n!) = \sum_{1 \leq x \leq n} \log x \approx \int_1^n \log x dx = n \log(n) + O(n)$$

The right hand side can be written as

$$\sum_{p \leq n} v_p(n!) \log p = \sum_{p \leq n} \log p \sum_{x \leq n} \sum_{a \geq 1, p^a | x} 1 = \sum_{p \leq n} \log p \sum_{a \geq 1} \sum_{x \leq n, p^a | x} 1$$

Which can be expressed as,

$$= \sum_{p \leq n} \log p \sum_{a \geq 1} \left\lfloor \frac{n}{p^a} \right\rfloor = \sum_{p \leq n} \log p \frac{n}{p} + O(n)$$

Finally we have,

$$\sum_{p \leq n} \log p \frac{n}{p} + O(n) = n \log(n) + O(n)$$

thus,

$$\sum_{p \leq n} \frac{\log p}{p} = \log(n) + O(1) \quad (5)$$

□



## 2 Sums of arithmetic functions

### 2.1 arithmetic functions

**Definition 2.1.** An arithmetic function is a complex-valued function on the positive integers,  $f : \mathbb{N}_{\geq 1} \rightarrow \mathbb{C}$ . We write  $\mathcal{A}$  for the  $\mathbb{C}$ -vector space of arithmetic functions.

The von Mangoldt function

$$\Lambda(n) = \begin{cases} \log p, & n = p^\alpha, \alpha \geq 1, \\ 0, & n \neq p^\alpha. \end{cases}$$

**Definition 2.2.** Let  $f$  be an arithmetic function. The summation function of  $f$  is the function defined on  $\mathbb{R}_{\geq 0}$  by

$$x \mapsto M_f(x) = \sum_{1 \leq n \leq x} f(n).$$

The summation function of  $f$  is a piecewise constant function, and in this chapter, we will present methods to study the following question:

### 2.2 Approximation by integrals

If  $f$  is the restriction to  $\mathbb{N}_{\geq 1}$  of a continuous function on  $\mathbb{R}$ , then  $M_f(x)$  is often well approximated by

$$\int_1^x f(t) dt.$$

For example, if  $f$  is *monotone* we have

**Theorem 2.3. Monotone comparison** If  $f$  is monotone we have

$$M_f(x) = \int_1^x f(t) dt + O(f(x) - f(1)). \quad (6)$$

One can use this result to estimate  $\log n!$ . Indeed, setting  $f(n) = \log n$ , one obtains

$$\log n! = \sum_{i=1}^n \log i = \int_1^n \log t dt + \Theta(\log n - \log 1) = n \log n - n + O(\log n).$$

thus providing a fairly good approximation for  $\log n!$ .

**Theorem 2.4** (Integration by parts). *Let  $g$  be an arithmetic function. Let  $a < b \in \mathbb{R}_{>0}$  and  $f : [a, b] \rightarrow \mathbb{C}$  a  $C^1((a, b])$  function. We have*

$$\begin{aligned} M_{fg}(b) - M_{fg}(a) &= \sum_{a < n \leq b} f(n)g(n) = [f(t)M_g(t)]_{t=a}^{t=b} - \int_a^b M_g(t)f'(t) dt \\ &= f(b)M_g(b) - f(a)M_g(a) - \int_a^b M_g(t)f'(t) dt. \end{aligned}$$

## 2.3 Dirichlet convolution

The Dirichlet convolution is a composition law on the set of arithmetic functions that realizes the multiplicative structure of the integers.

Let  $f, g \in \mathcal{A}$ , and define  $f * g \in \mathcal{A}$  by setting

$$(f * g)(n) = \sum_{ab=n} f(a)g(b) = \sum_{d|n} f(d)g(n/d).$$

**Example:**

$$\log = \Lambda * 1, \quad \text{i.e.} \quad \log(n) = \sum_{d|n} \Lambda(d).$$

Indeed, if  $n = \prod_p p^{\alpha_p}$  then

$$\begin{aligned} \log(n) &= \log\left(\prod_p p^{\alpha_p}\right) \\ &= \sum_p \alpha_p \log(p) \\ &= \sum_p \sum_{1 \leq \alpha \leq \alpha_p} \log(p) \\ &= \sum_{p^\alpha | n} \log(p) = \sum_{d|n} \Lambda(d). \end{aligned}$$

**Möbius inversion formula:** The *Möbius function* is by definition the inverse of the constant function 1:

$$\mu = 1^{(-1)}, \quad \mu(1) = 1, \quad \mu(n) = - \sum_{\substack{d|n \\ d < n}} \mu(d) \quad \text{for } n \geq 2.$$

In the following section we will show that

1. If  $n$  is divisible by a square not equal to 1 (i.e. there exists a prime  $p$  such that  $p^2 \mid n$ ), then  $\mu(n) = 0$ .
2. If  $n$  is square-free, and has  $r$  prime factors (i.e.  $n = p_1 \cdots p_r$ ), then  $\mu(n) = (-1)^r$ .

The inverse of the constant function 1 indicates that

$$\mu * 1 = \delta$$

$$\text{where } \delta(n) = \begin{cases} 1 & n = 1 \\ 0 & n > 1 \end{cases}$$

*Proof.* when  $n > 1$ ,

$$\begin{aligned} \mu * 1(n) &= \sum_{d \mid n} \mu(d) \\ &= \sum_{d \mid n, p^2 \mid d} \mu(d) + \sum_{d \mid n, p^2 \nmid d} \mu(d) \\ &= 0 + \sum_{d \mid n, p^2 \nmid d} \binom{r}{x} (-1)^x 1^{r-x} \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

□

## 2.4 Applications to Counting Prime Numbers

**Theorem 2.5** (Mertens). *We have*

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \log(x) + O(1). \quad (7)$$

*Proof.* start from

$$\sum_{n \leq x} \log n = x \log x + O(x) \quad (8)$$

The left hand side can be written as,

$$\sum_{n \leq x} \log n = \sum_{n \leq x} \sum_{d \mid n} \Lambda(d) = \sum_{d \leq x} \left\lfloor \frac{x}{d} \right\rfloor \Lambda(d) = \sum_{d \leq x} \frac{x}{d} \Lambda(d) + O(x)$$

Finally we have,

$$\sum_{d \leq x} \frac{x}{d} \Lambda(d) + O(x) = x \log x + O(x) \quad (9)$$

□

## 2.5 Multiplicative functions

**Definition 2.6.** A non-zero arithmetic function  $f$  is called multiplicative if and only if for all  $m, n \geq 1$  with  $(m, n) = 1$  we have  $f(mn) = f(m)f(n)$ . A non-zero arithmetic function is called completely multiplicative if for all  $m, n \geq 1$  we have  $f(mn) = f(m)f(n)$ .

**Proposition 2.7.** If  $f$  and  $g$  are multiplicative, then  $f * g$  and  $f^{(-1)}$  are as well.

### 3 Dirichlet Series

#### 3.1 Review of Power Series

For a sequence  $a_n$ , the power series is defined as,

$$F(a, q) = \sum_{n \geq 0} a_n q^n \quad (10)$$

The radius of convergence is defined as,

$$\frac{1}{\rho} = \limsup_{n \rightarrow \infty} |a_n|^{1/n} \quad (11)$$

Let  $b_n$  be another sequence with associated power series,

$$F(b, q) = \sum_{n \geq 0} b_n q^n$$

The product will be

$$F(a, q)F(b, q) = \sum_{n \geq 0} c_n q^n \quad c_n = \sum_{k+l=n} a_k b_l \quad (12)$$

#### 3.2 Dirichlet Series

Dirichlet series are to arithmetic functions as power series are to sequences of numbers.

Let  $f \in \mathcal{A}$  be an arithmetic function. The *Dirichlet series* associated to  $f$  is the series in the complex variable  $s$  given by

$$s \mapsto L(s, f) = \sum_{n \geq 1} \frac{f(n)}{n^s}.$$

**Definition 3.1.** An arithmetic function  $f : \mathbb{N}_{\geq 1} \rightarrow \mathbb{C}$  is of polynomial growth if it satisfies one of the following equivalent conditions.

- There exists a constant  $A \in \mathbb{R}$  (depending on  $f$ ) such that  $|f(n)| = O(n^A)$ .
- There exists  $\sigma \in \mathbb{R}$  such that the series  $L(\sigma, f)$  is absolutely convergent.

In this case we write

$$\sigma_f = \inf\{\sigma \in \mathbb{R} : L(\sigma, f) \text{ converges absolutely}\} \in \mathbb{R} \cup \{-\infty\};$$

The number  $\sigma_f$  is called the *abscissa of convergence* of  $L(s, f)$ .

*Proof.* Exercise. □

**Proposition 3.2.** *Let  $f$  be an arithmetic function with polynomial growth, and let  $\sigma_f$  be its abscissa of convergence. For all  $\sigma > \sigma_f$ , the series  $L(s, f)$  converges absolutely and uniformly in the half-plane  $\{s \in \mathbb{C} : \operatorname{Re}(s) \geq \sigma\}$ . In this domain, the derivative of  $L(s, f)$  is the Dirichlet series of the arithmetic function*

$$-\log f : n \mapsto -\log(n)f(n),$$

*that is to say,*

$$L'(s, f) = L(s, -\log f) = \sum_{n \geq 1} \frac{-\log(n)f(n)}{n^s},$$

*which has abscissa of convergence  $\sigma_f$  as well.*

*Proof.* To prove abscissa of convergence  $\sigma_{-\log \cdot f} = \sigma_f$ , for  $n \geq 3$ ,

$$\log n |f(n)| > |f(n)| \tag{13}$$

which indicates that  $\sigma_{-\log \cdot f} \geq \sigma_f$ . On the other hand,

$$L'(s, f) = L(s, -\log f) = \sum_{n \geq 1} \frac{-\log(n)f(n)}{n^s},$$

which converges on  $\operatorname{Re}(s) \geq \sigma_f$ , the function  $-\log \cdot f$  have a convergence area greater or equal than  $\operatorname{Re}(s) \geq \sigma_f$ , i.e.

$$\sigma_{-\log \cdot f} \leq \sigma_f$$

Thus we have  $\sigma_{-\log \cdot f} = \sigma_f$  □

The main reason to introduce Dirichlet series is the following.

**Theorem 3.3.** *Let  $f, g \in \mathcal{A}$ , with  $\sigma_f, \sigma_g < \infty$ . Then,  $\sigma_{f * g} \leq \max(\sigma_f, \sigma_g)$ , and for  $\operatorname{Re}(s) > \max(\sigma_f, \sigma_g)$  we have*

$$L(s, f * g) = L(s, f)L(s, g).$$

*Proof.* Let  $\operatorname{Re}(s) > \max(\sigma_f, \sigma_g)$ , so that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{|f * g(n)|}{|n^s|} &= \sum_{n=1}^{\infty} \frac{|\sum_{ab=n} f(a)g(b)|}{n^{\operatorname{Re}(s)}} \\ &\leq \sum_{n=1}^{\infty} \sum_{ab=n} \frac{|f(a)||g(b)|}{(ab)^{\operatorname{Re}(s)}} \end{aligned}$$

$$= \sum_{a,b=1}^{\infty} \frac{|f(a)||g(b)|}{(ab)^{\operatorname{Re}(s)}} = \left( \sum_{a=1}^{\infty} \frac{|f(a)|}{a^{\operatorname{Re}(s)}} \right) \left( \sum_{b=1}^{\infty} \frac{|g(b)|}{b^{\operatorname{Re}(s)}} \right) < \infty.$$

All of the above identities and swaps of order of summation above are justified by the fact that we are summing positive terms. We have thus shown that  $\sigma_{f*g} \leq \max(\sigma_f, \sigma_g)$ . Moreover, for  $\operatorname{Re}(s) > \max(\sigma_f, \sigma_g)$ , we have by absolute convergence that we can regroup the terms arbitrarily, and so we have

$$L(s, f)L(s, g) = L(s, f * g).$$

□

### 3.3 Dirichlet series and multiplicative functions

**Definition 3.4.** Let  $f \in \mathcal{A}$  be a multiplicative function of polynomial growth, then for all  $\sigma > \sigma_f$  we have

For all  $p$  prime, the series

$$L_p(s, f) := \sum_{a \geq 0} \frac{f(p^a)}{p^{as}}$$

converges absolutely and uniformly in the half plane  $\operatorname{Re}(s) \geq \sigma$ . We call  $L_p(s, f)$  the local factor of  $f$  at  $p$ .

**Theorem 3.5.** 1. Moreover, we have

$$L(s, f) = \prod_p L_p(s, f) = \lim_{P \rightarrow \infty} \prod_{p \leq P} L_p(s, f),$$

and the convergence is uniform in this half-plane.

2. More precisely, if we write

$$L^{>P}(s, f) = \prod_{p > P} L_p(s, f),$$

then as  $P \rightarrow \infty$  we have

$$L^{>P}(s, f) \rightarrow 1$$

uniformly in every half-plane  $\operatorname{Re}(s) \geq \sigma$ ,  $\sigma > \sigma_f$ .

3. Conversely, if  $f$  is an arithmetic function such that  $\sigma_f < \infty$  and  $f(1) = 1$  and if  $L(s, f)$  satisfies

$$L(s, f) = \prod_p L_p(s, f) = \lim_{P \rightarrow \infty} \prod_{p \leq P} L_p(s, f)$$

for  $s$  sufficiently large, then  $f$  is multiplicative.

*Proof.*

$$\begin{aligned}
 \prod_p L_p(s, f) &= \prod_p \sum_{a \geq 0} \frac{f(p^a)}{(p^a)^s} \\
 &= \sum_{a_1 \geq 0, a_2 \geq 0, \dots} \frac{f(p_1^{a_1}) f(p_2^{a_2}) \dots}{(p_1^{a_1})^s (p_2^{a_2})^s \dots} \\
 &= \sum_{a_1 \geq 0, a_2 \geq 0, \dots} \frac{f(p_1^{a_1} p_2^{a_2} \dots)}{(p_1^{a_1} p_2^{a_2} \dots)^s} \\
 &= \sum_{n \geq 1} \frac{f(n)}{n^s} = L(s, f)
 \end{aligned}$$

□

**Corollary 3.6.** *If  $f$  is completely multiplicative, then for  $\operatorname{Re}(s) > \sigma_f$  we have*

$$L(s, f) = \prod_p \left( 1 - \frac{f(p)}{p^s} \right)^{-1}.$$



## 4 Primes in Arithmetic Progressions

### 4.1 arithmetic progressions

**Definition 4.1.** *An arithmetic progression is a doubly-infinite **subset** of  $\mathbb{Z}$  satisfying the following property: There exists a positive integer  $q > 0$  such that the distance between two consecutive integers of this subset is always  $q$ . The integer  $q$  is called the modulus of the arithmetic progression.*

It is easy to see that arithmetic progressions of modulus  $q$  are of the form

$$L_{q,a} = a + q\mathbb{Z} \subseteq \mathbb{Z},$$

where  $a$  is an integer. We remark that if  $a \equiv a' \pmod{q}$ , then we have  $L_{q,a} = L_{q,a'}$ . Thus arithmetic progressions of modulus  $q$  are indexed by the congruence classes modulo  $q$  (i.e. by the ring  $\mathbb{Z}/q\mathbb{Z}$ ). There are therefore  $q$  of them. The integer  $a$  is called the class of the arithmetic progression.

**Theorem 4.2** (Dirichlet's theorem on primes in arithmetic progressions). *Let  $a, q > 0$  be two relatively prime integers. Then, the set*

$$\mathcal{P}_{q,a} = \mathcal{P} \cap L_{q,a}$$

*is infinite. Said differently, there exist infinitely many prime numbers  $p \equiv a \pmod{q}$ .*

In the vein of the prime number theorem, we can pose more precise questions on the density of the set  $\mathcal{P}_{q,a}$ . We therefore set

$$\pi(x; q, a) = |\mathcal{P}_{q,a} \cap [1, x]| = |\{p \leq x : p \equiv a \pmod{q}\}| = \sum_{\substack{p \equiv a \pmod{q} \\ p \leq x}} 1$$

the counting function of the primes  $p \equiv a \pmod{q}$ . At the beginning of the 20th century, Landau showed this generalization of the prime number theorem:

**Theorem 4.3.** (Landau). *Let  $a, q > 0$  be relatively prime integers. Then*

$$\pi(x; q, a) = \frac{1}{\varphi(q)} \pi(x) (1 + o(1)) = \frac{1}{\varphi(q)} \frac{x}{\log x} (1 + o(1)).$$

*where  $\varphi = \mu * \text{Id}$  is the Euler function,  $\mu = 1^{(-1)}$  is the Möbius function and  $\text{Id}(n) = n$*

is identity function.

Merten's theorem is extended to intersection set of primes and arithmetic progressions.

**Theorem 4.4.** *We have*

$$\sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \frac{\Lambda(n)}{n} = \frac{1}{\varphi(q)} \log(x) + O(1) \quad (4.1)$$

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod{q}}} \frac{\log p}{p} = \frac{1}{\varphi(q)} \log(x) + O(1), \quad (4.2)$$

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod{q}}} \frac{1}{p} = \frac{1}{\varphi(q)} \log \log(x) + O(1). \quad (4.3)$$

The crucial point is the group structure of the set  $(\mathbb{Z}/q\mathbb{Z})^\times$ .

## 4.2 abelian group

The Abelian group is the group in which the group operation is commutative, meaning that the order of operation does not affect the results.

**Definition 4.5.**  $(G, *)$  is called a group if it satisfies the following properties:

1. **Closure:** For all  $a, b \in G$ , we have  $a * b \in G$ .

2. **Associativity:** For all  $a, b, c \in G$ ,

$$(a * b) * c = a * (b * c).$$

3. **Identity element:** There exists an element  $e \in G$  such that

$$e * a = a * e = a, \quad \forall a \in G.$$

4. **Inverse element:** For every  $a \in G$ , there exists an element  $a^{-1} \in G$  such that

$$a * a^{-1} = a^{-1} * a = e.$$

If, in addition,  $a * b = b * a$  for all  $a, b \in G$ , then  $G$  is called an *abelian group*.

**Example:**

- $(\mathbb{Z}, +)$  is an abelian group.
- $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$  is an abelian group of addition modulo 2.

Let  $G$  be a finite abelian group, and  $g \in G$  is an element in the group. The action of right translation is defined as,

$$T_g f : g' \mapsto T_g f(g') = f(g'g) \quad (14)$$

We can verify that,

$$T_g \circ T_{g'} = T_{gg'} \quad (15)$$

As the action is invertible, the inverse is defined as,

$$(T_g)^{-1} = T_{g^{-1}} \quad (16)$$

As  $G$  is abelian group, the action can commute,

$$T_g \circ T_{g'} = T_{g'} \circ T_g \quad (17)$$

The inner product is defined as,

$$\langle f, f' \rangle = \frac{1}{|G|} \sum_{g \in G} f(g) \overline{f'(g)}. \quad (18)$$

An endomorphism is a structure-preserving map from a mathematical object to itself.

**Example:** For the group in  $(\mathbb{R}^2, +)$ , any  $2 \times 2$  matrix is an endomorphism.

**Definition 4.6** (Group homomorphism). *Let  $(G, \circ)$  and  $(H, *)$  be groups. A map  $\phi : G \rightarrow H$  is called a group homomorphism if*

$$\phi(x \circ y) = \phi(x) * \phi(y) \quad \text{for all } x, y \in G.$$

**Example:**  $\varphi(n) = n \bmod 5$ , which  $(\mathbb{Z}, +) \mapsto (\mathbb{Z}_5, +)$

**Definition 4.7** (Group character). *Let  $G$  be a group. A (group) character of  $G$  is a group homomorphism*

$$\chi : G \longrightarrow \mathbb{C}^\times,$$

*that is, a mapping into the group  $\mathbb{C}^\times$  with complex numbers set and the multiplication operation.*

**Example:** Let  $G = \mathbb{Z}_3 = \{0, 1, 2\}$  under addition modulo 3. A *character* of  $G$  is a homomorphism

$$\chi : (\mathbb{Z}_3, +) \longrightarrow (\mathbb{C}^\times, \cdot)$$

satisfying

$$\chi(a + b) = \chi(a)\chi(b), \quad \forall a, b \in \mathbb{Z}_3.$$

The characters of  $\mathbb{Z}_3$  are given by

$$\chi_k(a) = e^{2\pi i k a / 3}, \quad k = 0, 1, 2.$$

Each  $e^{2\pi i k a / 3}$  lies in  $\mathbb{C}^\times$  (since it is nonzero), and

$$\chi_k(a + b) = e^{2\pi i k (a+b) / 3} = e^{2\pi i k a / 3} e^{2\pi i k b / 3} = \chi_k(a)\chi_k(b).$$

Hence each  $\chi_k$  is a group homomorphism

$$\chi_k : (\mathbb{Z}_3, +) \rightarrow (\mathbb{C}^\times, \cdot).$$

**Theorem 4.8.** *Let  $(V, \langle \cdot, \cdot \rangle)$  be a finite-dimensional hermitian vector space, and  $\mathcal{T} \subset \text{End}(V)$  a set of pairwise commuting endomorphisms. Then there exists an orthonormal basis of  $V$  given by simultaneous eigenvectors of all of the elements of  $\mathcal{T}$  if and only if each  $T \in \mathcal{T}$  is normal (that is to say,  $TT^* = T^*T$ ).*

In the situation at hand, the spectral theorem implies that  $\mathcal{C}(G)$  possesses an orthonormal basis of eigenvectors of all of the  $T_g$ . In fact, up to permutation, this basis is unique.

**Theorem 4.9.** *There exists a unique orthonormal basis  $\hat{G}$  of  $\mathcal{C}(G)$  consisting of eigenvectors for all of the  $T_g$ , such that for all  $\chi \in \hat{G}$  we have  $\chi(e) = 1$ . We have an equality*

$$\hat{G} = \text{Hom}_{\mathbb{Z}}(G, \mathbb{C}^\times),$$

where  $\text{Hom}_{\mathbb{Z}}(G, \mathbb{C}^\times)$  designates the set of group homomorphisms from  $G$  to  $\mathbb{C}^\times$ . This set is a group: the group of characters of  $G$ , or also the dual of  $G$ .

**Example:** Fourier Transform defines a unique orthonormal basis  $\widehat{G}$ . For identity  $\delta[n]$ , its Fourier transform  $\mathcal{F}\{\delta[n]\} = 1$ .

**Proof.** The set  $\{T_g : g \in G\}$  is a family of unitary operators, so  $T_g^* = T_{g^{-1}}$ , and therefore normal and pairwise commuting. By the spectral theorem, this family is diagonalizable with respect to an orthonormal basis. Let  $\mathcal{B}$  be such a basis. The  $\psi \in \mathcal{B}$  are thus non-zero eigenvectors of all of the  $T_g$ , and we have for all  $g$  that

$$T_g \psi(x) = \psi(xg) = \chi_\psi(g) \psi(x). \quad (19)$$

In the case of  $\mathbb{Z}/q\mathbb{Z}$ , the character is defined as,

$$\psi_n(x) : x \mapsto e^{i2\pi nx/q} \quad x \in \mathbb{Z}/q\mathbb{Z} \quad (20)$$

Note that the character only depends on the class of  $n$  modulo  $q$ .

The trivial character is defined as,

$$\chi_0 : g \mapsto 1 \quad (21)$$

### 4.3 Dirichlet Character

**Definition 4.10** (4.11). Let  $q \geq 1$  be an integer. The characters of the abelian group  $(\mathbb{Z}/q\mathbb{Z})^\times$  are called Dirichlet characters of modulus  $q$ . The trivial character (i.e. the constant function equal to 1) is denoted  $\chi_0$ .

The multiplicative group  $(\mathbb{Z}/q\mathbb{Z})^\times$  is defined as the integer between 1 and  $q$  that is co-prime with  $q$  under multiplication operation. The Dirichlet character of this group is defined as group character *extended* to all integers,

$$\chi(n) = \begin{cases} \chi(n \bmod q), & (n, q) = 1, \\ 0, & (n, q) > 1. \end{cases}$$

which satisfies *completely multiplicative* property (by definition of group homomorphism),

$$\chi(mn) = \chi(m)\chi(n) \text{ for all } m, n \in N_{\geq}$$

We consider Dirichlet series associated with Dirichlet character,

$$L(\chi, s) = \sum_{n \geq 1} \frac{\chi(n)}{n^s} \quad (22)$$

which can be written as,

$$L(\chi, s) = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1} \quad (23)$$

**Proposition 4.11** (4.12). *Let  $\chi \pmod{q}$  be a Dirichlet character.*

- *If  $\chi = \chi_0$ , we have for  $\operatorname{Re}(s) > 1$*

$$L(s, \chi_0) = \prod_{p|q} \left(1 - \frac{1}{p^s}\right) \zeta(s),$$

*which admits an analytic continuation to the half-plane  $\operatorname{Re}(s) > 0$  with a simple pole at  $s = 1$ .*

- *If  $\chi \neq \chi_0$ , the series  $L(s, \chi)$  converges uniformly on compacta in the half-plane  $\operatorname{Re}(s) > 0$  and thus defines a holomorphic function in this domain. More precisely, we have for  $\operatorname{Re}(s) > 0$*

$$\sum_{1 \leq n \leq X} \frac{\chi(n)}{n^s} = L(s, \chi) + O\left(\frac{q|s|}{\sigma} X^{-\sigma}\right). \quad (4.12)$$

## 4.4 Beginning of the proof of Mertens theorem in arithmetic progressions

To prove Mertens theorem,

$$\sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \frac{\Lambda(n)}{n} = \frac{1}{\varphi(q)} \log(x) + O(1) \quad (4.1)$$

Consider the orthogonality, we have,

$$\delta_{\chi=\chi'} = \frac{1}{|G|} \sum_{g \in G} \chi(g) \overline{\chi'(g)} \quad (24)$$

$$\delta_{g=g'} = \frac{1}{\hat{G}} \sum_{\chi \in \hat{G}} \chi(g) \overline{\chi(g')} \quad (25)$$

which are dual relations. For the group  $(\mathbb{Z}/q\mathbb{Z})^\times$ , we have

$$\delta_{\chi=\chi'} = \frac{1}{\varphi(q)} \sum_{a \bmod q} \chi(a) \overline{\chi'(a)} \quad (26)$$

$$\delta_{a \equiv b \bmod q} = \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \chi(a) \overline{\chi(b)} \quad (27)$$

where  $\varphi(q)$  is the Euler function.

we have

$$\sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \frac{\Lambda(n)}{n} = \frac{1}{\varphi(q)} \sum_{\chi \pmod{q}} \overline{\chi}(a) S_\chi(x), \quad (28)$$

where

$$S_\chi(x) = \sum_{n \leq x} \frac{\chi(n)}{n} \Lambda(n).$$

If  $\chi = \chi_0$ ,

$$S_{\chi_0}(x) = \log x + O_q(1)$$

It suffices to prove that the remain terms,

$$S_\chi(x) = O_q(1)$$

The sum of  $\chi \neq \chi_0$  can be written as,

$$\sum_{g \in G} \chi(g) = 0$$

as it is periodic,

$$|M_\chi(x)| = \left| \sum_{n \leq x} \chi(n) \right| \leq q$$

For the function,

$$T_\chi(x) = \sum_{n \leq x} \frac{\chi(n)}{n} \log(n)$$

by integral of parts

$$T_\chi(x) = \frac{\log(x)}{x} M_\chi(x) + \int_1^x M_\chi(x) (\log n - 1) \frac{1}{n^2} dn$$

is bounded by,

$$T_\chi(x) = O_q(1)$$

On the other hand,  $\log = 1 * \Lambda$ ,

$$\begin{aligned}
T_\chi(x) &= \sum_{n \leq x} \frac{\chi(n)}{n} \sum_{ab=n} \Lambda(a) \\
&= \sum_{n \leq x} \frac{\chi(ab)}{ab} \sum_{ab=n} \Lambda(a) \\
&= \sum_{a \leq x} \frac{\chi(a)}{a} \Lambda(a) \sum_{b \leq x/a} \frac{\chi(b)}{b} \\
&= S_\chi(x) L(\chi, 1)
\end{aligned}$$

As  $L(\chi, 1) \neq 0$ , we can prove that,

$$S_\chi(x) = O_q(1)$$